SPY Underlying Asset Arbitrage:

Implementation:

- **By SPY when see buy MSFT, sell when see SELL**
- **Implement our R_under calculation, capitalize price difference**
- **Use z-score instead of direct price difference**
- **Calculate price using top of book as opposed to last trade price**
- **Instead of target position size, just use order size**
- **Dynamic weighting of trade size**
- **Change from market order to market with leftover as limit**
- **Parsing more tickers for larger %, use weights from 2024, not now**

Since we don't have all underlying assets and official market cap doesn't update Realtime, it would be best to stick with factsheet data that updates quarterly. These three equations will make up the backbone of this (the numbers here are based off the factsheet of 9/30/2025:

$$R_{under} = \sum w_i P_i$$

$$= (9.88\%) * (NVDA) + (8.39\%) * (MSFT) + (5.10\%) * (AMZN) + (3.08\% + 2.88\%) * (GOOG)$$

$$R_{QQQ} = \sum (w_i) * P_{QQQ}$$

$$= \big(9.88\% + 8.39\% + 5.10\% + (3.08\% + 2.88\%)\big) * (QQQ)$$

$$\Delta = R_{under} - R_{QQQ}$$

$$\Delta^{+} = long\ signal, \qquad \Delta^{-} = short\ signal$$

W_i = weight of stock

P_i = Intraday return of the stock

P_{QQQ} = Intraday return of QQQ

R_{under} = R-Score of underlying assets

R_{QQQ} = R-Score of QQQ

Delta = R-score differential

The first equation tracking movement of the underlying asset region, by the summation of multiplying the weight of the underlying asset and multiplying it by the intraday return of said asset. The second equation is utilizing the combined weight of those underlying assets that are being tracked and multiplying it by the intraday return of QQQ. Then a difference between these two values is calculated, and a delta positive means that the return of the underlying assets is higher than the return of QQQ based of the weight of these stocks, which is an indication of a long signal. A delta negative indicates that the return of QQQ is higher than the return of the underlying assets, which is an indication of a short signal. A theoretical delta zero means that there is no arbitrage signal and the trade window is closed. While we can trade based of those indicators, it may not be successful since it does not account for trading error and noise.

$$Z = \frac{\Delta}{\sigma}$$

Sigma = Rolling Standard Deviation of Delta values

The value Z may be more reliable in this case. Sigma is the standard deviation of the delta, and this would help with avoiding trading sequences when the asset is noisy. Finding thresholds that Z trades best at would come with trial and error, but I think anything below 3 should not be traded, so that's a good start there.

Another thing that can be useful is to understand the volume of these underlying assets to determine the strength of the potential movement.

$$E = \frac{V_{under,avg}}{V_{QQQ}} = \frac{\frac{\sum V_i M_i w_i}{\sum M_i}}{V_{QQQ} \sum w_i}$$

$$E = \frac{V_{under,avg}}{V_{QQQ}} = \frac{\sum V_i w_i}{V_{QQQ} \sum w_i}$$

E = E-score

V_i = volume of stock

M_i = Market cap of stock

E will calculate this. The underlying asset average volume is market cap focused and divided the volume of QQQ. There should be a certain ratio that trading should be considered acceptable and unacceptable. We can play around with that later, but a good

starting point could be 1, meaning that the volume average of the underlying stocks exceeds the volume of QQQ.

SPX/SPY Direct Arbitrage:

In order to understand the relationship between SPY and SPX, we must understand the divisor:

$$D = \frac{S_{SPY}}{\sum N_{i,SPY}}$$

D = 10*

$$P_{SPX,D} = P_{SPX}/D$$

D = Divisor

S = Shares Outstanding

N = shares of stock held by SPY

$P_{SPX,D}$ = Price of SPX reflected at SPY scale

This is how we can model an SPX share price at the scale of SPY for arbitrage reasons. D is supposed to be around a theoretical 10, but there are variations, so optimizing this equation will be important to complete successful arbitrage trades. However, the same ideas fall here like the last strategy.

$$\Delta = P_{SPY} - P_{SPX,D}$$

$$\Delta^{+} = short\ signal, \qquad \Delta^{-} = long\ signal$$

However, this time the delta values are flipped and here is why. Since SPY is the tracking ETF to SPX, it is going to move in the direction of SPX. So if SPY is richer, it produces a positive signal, but it means there is expectation of it to go down.

$$Z = \frac{\Delta}{\sigma}$$

The Z-score equation will also be applied here, however there is another z-score that I would also like to apply to this strategy as well as the last strategy.

$$(z) = \frac{C}{\sigma}$$

C = Bid-ask spread + fees + slippage

The sigma is the same here. However, this other z score will be compared with the original z-score in order to to determine if the trade is acceptable.

$$Z > (z)\ , Short$$

$$Z < -(z)\ , Long$$

In this case, this is the logic that will be used. However, the logic is flipped for the first strategy since the deltas are flipped in signals. Another thing to keep in mind is that another trade condition is that delta must be greater than the bid-ask spread + fees, and the stop condition is when the delta is equal to the bid-ask spread + fees.